

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 18

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Natural processes may concentrate the material (eg adsorption onto sediments, bioaccumulation and food-chain concentration) • It may be carried to locations where it is hazardous, such as fishing grounds or beaches used by people	2
01.2	<i>Any 4 from:</i> • Physical state of the effluent (solution / solid) • Ocean currents / river flow • Effect of pH and oxygen on solubility • Bioaccumulation and food-chain concentration • Food sources and consumption by local people (eg shellfish, seaweed) • Half-lives of the isotopes • Meteorological conditions for any gaseous release • Nature of the environment - geology, vegetation	4
01.3	• Monitoring focuses on the sites most at risk of contamination • No need to sample sites that were never likely to be polluted; sampling checks the accuracy of the predictions	2
01.4	• It assesses the risk to the people most exposed because of their lifestyle before any health effects occur • If their risk is acceptably low everyone else has a lower risk; if unacceptable, emissions are reduced to prevent future exposure	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
02.1	• Solar PV cost fell dramatically from about 480 to 50 \$/MWh (about 90%), the steepest fall • Wind fell steadily from 200 to 45 • Gas stayed roughly constant at 60-70; solar and wind became cheaper than gas around 2015-2020 and are projected to fall further	3

Question	Marking guidance	Marks
02.2	• Development / research costs are still being paid for and there are no economies of scale in manufacture • They compete with established technologies whose development costs were paid long ago and which benefit from mass production and existing infrastructure • As output grows, mass production, improved designs / efficiency and experience reduce costs	3
02.3	• Most of the renewable cost is the initial equipment, paid for with loans on which interest must be paid; running costs are very low • Most of the gas project's cost is running costs (fuel) paid from income without borrowing, so it has lower interest payments	2
02.4	• Environmental damage (acid rain damage to buildings, crops, forests; health effects of air pollution; climate change) is paid for by others (health service, agriculture) not the energy user • Some impacts (climate change) cannot yet be costed; decommissioning and waste costs are also uncertain	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 179-181; AQA spec 3.3.2-3.3.3.	
03.1	• Gravimetry: gravimeters detect variations in gravity caused by differences in rock density / mass (metal ores usually denser than sedimentary rock) • Magnetometry • Seismic survey: sound / shock waves from controlled explosions or a seismic vibrator; echoes recorded by geophones give the depth, density and shape of rock strata • Resistivity	4
03.2	• Trial drilling is the most expensive technique per site • Remote sensing (satellite, aircraft, ground surveys) covers large areas at low cost per unit area and identifies promising sites • Drilling is then targeted where deposits are likely; it is the only method that produces actual rock samples for chemical analysis to confirm composition and purity (proven reserves)	3
03.3	• Different minerals emit / reflect infrared at different wavelengths • Multispectral sensors detect these wavelengths over the whole surface (eg Landsat 8 every 16 days) giving information on surface mineral composition	2
03.4	• Results are available immediately in the field, saving time waiting for laboratory analysis	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 137; AQA spec 3.2.3.4.	
04.1	• Radiation absorbed by tissue produces free radicals (ions with unpaired electrons), mostly from water, which are highly reactive • Free radicals damage DNA / chromosomes in the nucleus causing mutations (cancer, inherited effects) - rapidly dividing cells (skin, gut, bone marrow) most affected • Damage to other cell parts can stop the cell functioning or kill it; low damage may be repaired by the intact nucleus	3

Question	Marking guidance	Marks
04.2	- Acute: effects appearing quickly after a large dose over a short time (radiation sickness, damage to bone marrow / immune system / gut, haemorrhage, rapid death) - Chronic: effects appearing slowly, in proportion to the dose received over a long period (cancers, cataracts)	2
04.3	- Exposure: absorption of ionising radiation from a source the person is close enough to - Contamination: physically carrying radioactive material on or in the body, so exposure continues wherever the person goes; sealed sources can expose but not contaminate	2
04.4	- Sieverts = grays x radiation weighting factor (20 for alpha) $0.002 \times 20 = 0.04 \text{ Sv}$ (40 mSv)	2
04.5	The activity of a source (number of decays per second)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 301, 303; AQA spec 3.4.3.2.14.	
05.1	$(7.7 - 3.9) / 7.7 \times 100$ = 49.4% (accept 49%)	2
05.2	- Floating sea ice has already displaced its own mass of water; when it melts it contracts and occupies the same volume as the ice below the water line - Melting land ice (glaciers, ice sheets) adds new water to the sea, increasing its volume	2
05.3	- Ice and snow have a high albedo and reflect most sunlight; open sea water has a low albedo - As ice area shrinks more sunlight is absorbed by the sea and converted to heat, raising temperatures - This melts more ice: a positive feedback mechanism; thinner ice melts more easily each summer	3
05.4	- Thermal expansion: warmer sea water expands (slow because of the huge volume and high heat capacity; only surface water warms directly) - Melting of land ice (glaciers, Greenland and Antarctic ice sheets) adds water to the oceans	2
05.5	Ice shelves grounded on the seabed hold back / block the forward movement of the glaciers or ice sheet behind them; once removed the land ice flows faster (eg West Antarctica)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.	
06.1	B - tiny pores between clay particles allow capillary rise; none in sand	1
06.2	- Hygroscopic: held tightly by adhesive forces on particle surfaces; cannot drain and cannot be absorbed by plants - Capillary: held in pores by surface tension; does not drain but can be absorbed by plants - Gravitational: drains freely through large pores; available to plants only briefly as it passes	3
06.3	Small round crumbs leave spaces giving good drainage, aeration and easy root penetration; large flat platy peds pack together reducing all three	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	

Question	Marking guidance	Marks
07.1	- Human = $120 + 60 + 30 = 210$; total = 325 $210 / 325 \times 100 = 64.6\%$ (accept 65%)	2
07.2	- Nitrogen and hydrogen react to form ammonia over an iron catalyst at high temperature and pressure; ammonia then converted to nitrates for fertiliser - Uses large amounts of energy from fossil fuels (also the hydrogen source), releasing CO₂ and NO_x - Biological fixation by bacteria occurs at low temperature using solar energy captured by plants (enzymes lower activation energy)	3
07.3	Any 3 from: - Eutrophication of water bodies by leached nitrates / sewage - Photochemical smogs and tropospheric ozone from NO_x - Acid rain (nitric acid) - Nitrous oxide as a greenhouse gas - NO_x toxicity / respiratory disease - Nitrate in drinking water (methaemoglobinaemia)	3
07.4	- Carbon is lost as CO₂ (composting) or CH₄ (digestion) so the C:N ratio improves / more nitrogen-rich - Bulk and odour reduced, pathogens killed, easier to apply; digestion also yields biogas	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 154-155; AQA spec 3.2.4.3.	
08.1	- Deforestation: less interception so less evaporation from leaves; less transpiration so less water returned to the atmosphere and less precipitation downwind; more infiltration and surface runoff, higher river flows / flooding - Urban development: impermeable concrete / tarmac reduce infiltration and aquifer recharge; runoff is faster and greater, increasing river flooding downstream; drains carry water quickly to rivers - Agriculture: soil compaction by machinery / trampling reduces infiltration and soil moisture, increases runoff; irrigation increases evaporation; loss of soil biota (worms) reduces infiltration and water retention; drainage lowers water tables	6
08.2	Any 2 from: - Higher temperatures increase evaporation and change the amount, timing, type (rain vs snow) and location of precipitation - More rapid melting of snow and ice changes river regimes: more flooding after rain, lower flows in dry periods - Changed wind patterns move humid air to different areas - Sea level rise / salt incursion into coastal aquifers	2
08.3	- Most water abstracted is cleaned and returned to the river / sea so the main impact is on quality (pollution) not quantity - Effects are noticeable only where large urban centres draw on restricted local supplies (aquifers, small rivers)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 116-117; AQA spec 3.2.2.1.	
09.1	- t-test - The data are measured (continuous) values and two means are being compared - The data are (assumed to be) normally distributed	3

Question	Marking guidance	Marks
09.2	• Use the Mann-Whitney U test instead (compares medians)	1
09.3	• There is a 2% probability that the difference occurred by random chance • As this is less than 0.05 the difference is statistically significant / the null hypothesis is rejected	2
09.4	• The ranges of values (mean plus or minus SD) do not overlap: woodland up to 1.6 and clearing down to 2.4 • About 68% of values lie within one SD of each mean so the two data sets are clearly separated	2
09.5	• Significance shows the difference is unlikely to be due to chance, not that a causal relationship exists • Another variable (eg topography / aspect of the sites) could cause both; a controlled experiment or further evidence is needed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Smoke: solid particulates (PM10, PM5, PM1) from incomplete combustion; small particles stay suspended, penetrate lungs; toxic chemicals (acids, phenols, metals) carried on particles; high albedo cools climate; blocks light reducing photosynthesis; damages buildings • Smoke smogs form under temperature inversions in valleys (London 1952, 12 000 deaths); synergism with SO₂ • Controls: Clean Air Act 1956 (smokeless zones), smokeless fuels / gas / electricity, coal treatment, electrostatic precipitators, cyclones, bag filters, scrubbers, DPFs, better combustion; smoke smogs have largely been eliminated in UK cities • Acid rain gases: SO₂ (coal combustion, smelting sulfide ores) oxidised by ozone to SO₃ - sulfuric acid; NO_x (high temperature combustion) - nitric acid; HCl from PVC incineration; reactive gases travel long distances (transboundary pollution) • Effects: pH denatures proteins / enzymes in exposed cells (gills, eggs, root hairs, stomata); dissolves exoskeletons; leaches Ca and Mg then mobilises toxic Al and Pb; corrodes metals, dissolves limestone; severity depends on bedrock (limestone neutralises) • Controls: fuel desulfurisation (coal washing, hydrodesulfurisation, low-sulfur ship fuels), dry and wet FGD, fluidised bed combustion, catalytic converters, urea sprays; switching from coal to gas / renewables • Evaluation: SO₂ emissions in Europe have fallen dramatically and lake pH is recovering; NO_x harder to control because of growing vehicle numbers; catalytic converters need lead-free fuel and warm engines; FGD is costly and produces wastes / gypsum; controlling primary pollutants also prevents secondary pollutants (ozone, PANs) • Judgement on relative success of controls for particulates vs SO₂ vs NO_x	9
10.2	<i>Any 1 from:</i> • Reduces growth rates of trees / crops • Damages leaf cuticles (allowing SO₂ to cause more damage)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-254; AQA spec 3.4.3.2.1-3.4.3.2.5.	

Question	Marking guidance	Marks
11.1	Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Nature of the problem: interconnected atmosphere-hydrosphere-biosphere systems; changes over different time and spatial scales; natural fluctuations (solar output, orbit) masking or exaggerating anthropogenic change; time delay between cause and effect (ocean heat capacity) • Limitations of historic data: sparse global coverage, unsophisticated equipment, heat island bias, no atmospheric composition records • Proxy data and their reliability: dendrochronology, coral growth rings, pollen in dated sediments, ice cores (gas bubbles, isotope ratios, volcanic ash layers, compaction problems) • Modern monitoring: long-term data sets (Mauna Loa), electronic equipment, satellites (LEO vs geostationary), Argo floats, remote sensing of ice thickness / area / sea level • Feedback mechanisms and tipping points adding uncertainty; ecological uncertainty (species survival, distribution, fragmentation) • Computer models: how they are tested and refined; importance of accurate representative data; scenario-based predictions and ranges • Uncertainty over ENSO / Conveyor / jet stream responses • Evaluation and conclusion: confidence in the overall trend and human cause despite detailed uncertainty; role of peer review and international bodies (IPCC) Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90, 100-104, 10; AQA spec 3.2.1.2, 3.1.1.3.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Scale of the problem: how energy is used (transport, buildings, industry, water); links between affluence and per capita use • Transport: bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • Buildings: orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • Industry: heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • Limits: rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples; comparison with the cost of new supply; conclusion that conservation is the cheapest first step but cannot alone meet demand growth Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	25